

Mathematics Section A

1. Let $[\cdot]$ denote the greatest integer function. If the domain of the function $f(x) = \cos^{-1}\left(\frac{4x+2[x]}{3}\right)$ is $[\alpha, \beta]$, then $12(\alpha + \beta)$ is equal to:
(A) 6 (B) 8 (C) 9 (D) 4
2. If the set of all solutions of $|x^2 + x - 9| = |x| + |x^2 - 9|$ is $[\alpha, \beta] \cup [\gamma, \infty)$, then $(\alpha^2 + \beta^2 + \gamma^2)$ is equal to:
(A) 9 (B) 18 (C) 36 (D) 72
3. Let z be a complex number such that $|z + 2| = |z - 2|$ and $\arg\left(\frac{z+3}{z-i}\right) = \frac{\pi}{4}$. Then $|z|^2$ is equal to:
(A) 9 (B) 4 (C) 5 (D) 1
4. The number of functions $f: \{1, 2, 3, 4\} \rightarrow \{a, b, c\}$, which are not onto, is:
(A) 48 (B) 45 (C) 51 (D) 35
5. Let $S = \{A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \in \{0, 1, 2, 3, 4\} \text{ and } A^2 - 4A + 3I = 0\}$
(A) 20 (B) 17 (C) 21 (D) 19
6. Let $A = \begin{bmatrix} 1 & 1 & 2 \\ -2 & 0 & 1 \\ 1 & 3 & 5 \end{bmatrix}$. Then the sum of all elements of the matrix
(A) 3 (B) 4 (C) -4 (D) -3
7. The first term of an A.P. of 30 non-negative terms is $\frac{10}{3}$. If the sum of this A.P. is the cube of its last term, then its common difference is:
(A) $\frac{5}{87}$ (B) $\frac{25}{83}$
(C) $\frac{15}{29}$ (D) $\frac{5}{29}$
8. The number of ways, of forming a queue of 4 boys and 3 girls such that all the girls are not together, is:
(A) 5040 (B) 3050 (C) 3410 (D) 4320
9. Let the smallest value of $k \in \mathbb{N}$, for which the coefficient of x^3 in $(1+x)^3 + (1+x)^4 + (1+x)^5 + \dots + (1+x)^{99} + (1+kx)^{100}$, $x \neq 0$, is $(43n + \frac{101}{4})^{(100C_3)}$ for some $n \in \mathbb{N}$, be p . Then the value of $p + n$ is:
(A) 10 (B) 11 (C) 12 (D) 13
10. Suppose that the mean and median of the non-negative numbers 21, 8, 17, a , 51, 103, b , 13, 67, ($a > b$), are 40 and 21, respectively. If the mean deviation about the median is 26, then $2a$ is equal to:
(A) 109 (B) 117 (C) 161 (D) 131
11. Let the line $L_1: x + 3 = 0$ intersect the lines $L_2: x - y = 0$ and $L_3: 3x + y = 0$ at the points A and B, respectively. Let the bisector of the obtuse angle between the lines L_2 and L_3 intersect the line L_1 at the point C. The $BC^2 : AC^2$ is equal to:
(A) 5 : 1 (B) 1 : 5 (C) 2 : 3 (D) 3 : 2
12. Let the vertex A of a triangle ABC be (1, 2), and the mid-point of the side AB be (5, -1). If the centroid of this triangle is (3, 4) and its circumcenter is (α, β) , then $21(\alpha + \beta)$ is equal to:
(A) 309 (B) 403 (C) 497 (D) 524
13. Suppose that two chords, drawn from the point (1, 2) on the circle $x^2 + y^2 + x - 3y = 0$ are bisected by the y-axis. If the other ends of these chords are R and S, and the mid point of the line segment RS is (α, β) , then $6(\alpha + \beta)$ is equal to:
(A) 1 (B) 3 (C) 4 (D) 6
14. A line with direction ratios 1, -1, 2 intersects the line $\frac{x}{2} = \frac{y}{3} = \frac{z+1}{3}$ and $\frac{x+1}{-1} = \frac{y-2}{1} = \frac{z}{4}$ at the points P and Q, respectively. If the length of the line segment PQ is α , then $225\alpha^2$ is equal to:
(A) 1024 (B) 1014 (C) 1104 (D) 1204
15. The square of the distance of the point (-2, -8, 6) from the line $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z}{-1}$ along the line $\frac{x+5}{1} = \frac{y+5}{-1} = \frac{z}{2}$ is equal to:
(A) 3 (B) 6 (C) 8 (D) 12
16. If $y = \tan^{-1}\left(\frac{3\cos x - 4\sin x}{4\cos x + 3\sin x}\right) + 2\tan^{-1}\left(\frac{x}{1+\sqrt{1-x^2}}\right)$, then $\frac{dy}{dx}$ at $x = \frac{\sqrt{3}}{2}$ is equal to:
(A) 3 (B) -1 (C) 1 (D) 2
17. Let f be a real polynomial of degree n such that $f(x) = f'(x)f''(x)$, for all $x \in \mathbb{R}$. If $f(0) = 0$, then $36(f'(2) + f''(2) + \int_0^2 f(x) dx)$ is equal to:
(A) 42 (B) 46 (C) 56 (D) 66
18. The area of the region $\{(x, y) : y \leq \pi - |x|, y \leq |x \sin x|, y \geq 0\}$ is:

- (A) $1 + \frac{\pi^2}{8}$ (B) $2 + \frac{\pi^2}{4}$
 (C) $\frac{\pi^2}{8} - 1$ (D) $4 + \frac{\pi^2}{2}$
19. Let $\int_{-2}^2 (|\sin x| + [x \sin x]) dx = 2(3 - \cos 2) + \beta$, where $[.]$ is the greatest integer function. Then $\beta \sin(\frac{\beta}{2})$ equals:
 (A) 1 (B) 2 (C) 4 (D) 8
20. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = (1 + x + x^2)(1 - y + y^2)$, $y(0) = \frac{1}{2}$. Then $(2y(1) - 1)$ is equal to

- (A) $\sqrt{3} \tan(\frac{11\sqrt{3}}{6})$
 (B) $\frac{\sqrt{3}}{2} \tan(\frac{11\sqrt{3}}{12})$
 (C) $\sqrt{3} \tan(\frac{11\sqrt{3}}{12})$
 (D) $\frac{\sqrt{3}}{2} \tan(\frac{11\sqrt{3}}{6})$

Mathematics Section B

21. A coin is tossed 8 times. If the probability that exactly 4 heads appear in the first six tosses and exactly 3 heads appear in the last five tosses is p , then $96p$ is equal to ____.
22. Consider the parabola $P : y^2 = 4kx$ and the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Let the line segment joining the points of intersection of P and E , be their latus rectums. If the eccentricity of E is e , then $e^2 + 2\sqrt{2}$ is equal to ____.
23. If $A = \frac{\sin 3^\circ}{\cos 9^\circ} + \frac{\sin 9^\circ}{\cos 27^\circ} + \frac{\sin 27^\circ}{\cos 81^\circ}$ and $B = \tan 81^\circ - \tan 3^\circ$, then $\frac{B}{A}$ is equal to ____.
24. Let $\vec{a}_k = (\tan \theta_k) \hat{i} + \hat{j}$ and $\vec{b}_k = \hat{i} - (\cot \theta_k) \hat{j}$, where $\theta_k = \frac{2^{k-1}\pi}{2^n+1}$, for some $n \in \mathbb{N}, n > 5$. Then the value of $\frac{\sum_{k=1}^n |\vec{a}_k|^2}{\sum_{k=1}^n |\vec{b}_k|^2}$ is
25. The number of points, at which the function $f(x) = \max\{6x, 2 + 3x^2\} + |x - 1| \cos|x^2 - \frac{1}{4}|$, $x \in (-\pi, \pi)$, is not differentiable, is ____.

Physics Section A

26. In a screw gauge when the circular scale is given five complete rotations it moves linearly by 2.5 mm. If the circular scale has 100 divisions, the least count of screw gauge is ____ mm.
 (A) 1×10^{-2} (B) 1×10^{-3}
 (C) 5×10^{-2} (D) 5×10^{-3}
27. The increase in the pressure required to decrease the volume (ΔV) of water is $6.3 \times 10^7 \text{ N/m}^2$. The percentage decrease in the volume is _____. (Bulk modulus of water = $2.1 \times 10^9 \text{ N/m}^2$.)
 (A) 2% (B) 3% (C) 6% (D) 4%
28. The time taken by a block of mass m to slide down from the highest point to the lowest point on a rough inclined plane is 50% more compared to the time taken by the same block on identical inclined smooth plane. Both inclined planes are at 45° with the horizontal. The coefficient of kinetic friction between the rough inclined surface and block is _____.
 (A) $3/4$ (B) $2/3$ (C) $5/9$ (D) $4/9$
29. Two nuclei of mass number 3 combine with another nucleus of mass number 4 to yield a nucleus of mass number 10. If the binding energy per nucleon for the mass numbers 3, 4 and 10 are 5.6 MeV, 7.4 MeV and 6.1 MeV respectively, then in the process, $\Delta Mc^2 =$ ____ MeV.
 (A) 6.9 (B) 7.9 (C) 2.2 (D) 4.3
30. A solid sphere of mass M and radius R is divided into two unequal parts. The smaller part having mass $M/8$ is converted into a sphere of radius r and the larger part is converted into a circular disc of thickness t and radius $2l$. If I_1 is moment of inertia of a sphere having radius about an axis through its centre and I_2 is the moment of inertia of a disc about its diameter, the ratio of their moment of inertia $I_2/I_1 =$ _____.
 (A) 35 (B) 70 (C) 140 (D) 210
31. The two projectiles are projected with the same initial velocities at the 15° and 30° with respect to the horizontal. The ratio of their ranges is $1 : x$. The value of x is
 (A) $\sqrt{2}$ (B) $\sqrt{3}$
 (C) $2\sqrt{3}$ (D) $\frac{1}{\sqrt{2}}$